MUHAMAD FAIZ BIN IBRAHIM

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Portfolio: https://mfaizzz.github.io/

SUMMARY

Data Engineer with a foundation in Information Systems and Data Science, experienced in end-to-end cloud pipelines, enterprise system integration and analytics delivery. Specialises in Snowflake, Azure Data Factory, Microsoft Fabric and SQL/PySpark ELT across multi-region environments. Current independent EFTECH engagement supports a PETRONAS Data Security project integrating 15 source systems. Combines data platform engineering with data quality, governance, security reporting and cross-functional delivery.

EXPERIENCE

Domino's Pizza Enterprises Sdn Bhd, ***Data Engineer***

May 2025 - Present

- Designed, developed, and maintained end-to-end ETL/ELT pipelines using Azure Data Factory and Snowflake, supporting secure data ingestion, transformation, incremental loading, deduplication, and operational reliability for downstream reporting and analytics.
- Built and maintained Snowflake objects including tables, file formats, external stages, and stored procedures across RAW, ODS, MDL, and SHARED layers to support scalable data processing.
- Developed and enhanced Azure Data Factory pipelines using parameterized activities, For Each loop, dynamic file filtering, and Snowflake script execution for reusable and efficient orchestration.
- Worked on data warehouse pipelines and external API integrations, including PayPal API integration, to streamline data flow, improve reporting accuracy, and support enterprise analytics.
- Optimized pipeline and query performance through partition pruning, CTE refactoring, deduplication strategies, and Snowflake warehouse tuning to improve stability and efficiency.
- Applied data governance and quality practices through data profiling, tagging, validation, and metadata alignment across schema layers.
- Managed environment configuration, secrets, and deployment support across CI, SBX, Stage, and Prod using Terraform and Azure platform services.
- Collaborated on change requests, pull requests, code reviews, testing, and production rollouts using Azure DevOps and Git.
- Investigated and resolved pipeline, data quality, and production issues by tracing data across source systems, ADF workflows, Snowflake layers, and downstream outputs.

- Contributed to PayPal Germany and Netherlands SFTP-to-Azure Blob-to-Snowflake RAW/ODS reconciliation feeds, data-quality parity and legacy-feed migration.
- Contributed to Adobe Learning Manager and Captivate certification reporting for IDAM; supported Tanda and JP Kintai workforce ingestion and Goodays/Critzr store-feed corrections.
- Implemented and supported managed-identity and Snowflake key-pair authentication changes across ADF and SSIS; contributed to FIN Navigator and Yext Azure feed modernisation.
- Developed FoodPanda Taiwan Blob/Event Grid/Snowpipe RAW/ODS ingestion in DEV/acceptance, with release preparation and operational documentation.
- Additional involvement: Adyen, GMO, RAT, Stats NZ, PULSE/OLO, Taguchi, Nexus/IDAC, Turnkey e-invoicing, Uber Eats ingestion, store/stock feeds, and FX/DIME/SSIS migration assessments.

EFTECH Sdn Bhd, ***Independent Data Engineer - Freelance (Client: PETRONAS)***

May 2026 - Present (Concurrent Engagement)

- Develop Microsoft Fabric data integration and Power BI dashboards for a PETRONAS Data Security project spanning 15 source systems, concurrently with the main DPE role.
- Build Fabric/PySpark transformations and Delta processing to consolidate security issue logs, reconcile source records and align system/table metadata for analytics.
- Develop Power BI and DAX security-score, classification and remediation reporting; validate duplicate records, reporting mappings and metric definitions.
- Collaborate on data governance and security reporting requirements to translate multi-system data into clear dashboards for stakeholders.

EFTECH Sdn Bhd, ***Senior Data Analyst***

Nov 2024 - Feb 2025

- Developed and optimized interactive Power BI dashboards for PETRONAS Upstream by integrating data from SQL Server and Oracle sources to support business and operational reporting.
- Contributed to the design, enhancement, and maintenance of key reporting solutions, including the Upstream Data Control Dashboard, Upstream Data Security Dashboard (UDSec), and Upstream Data Assurance Index.
- Automated recurring reporting and monitoring workflows using Power Automate, reducing manual effort, improving turnaround time, and increasing overall process efficiency.
- Performed data cleansing, validation, and reconciliation to ensure high data accuracy, consistency, and compliance with internal reporting and governance standards.
- Worked closely with stakeholders to translate business requirements into actionable dashboard features, improving visibility into data governance, security, control, and assurance metrics across PETRONAS Upstream.
- Supported end-to-end dashboard development, from data extraction and SQL integration to visualization design, performance optimization, and report delivery.
- Strengthened reporting reliability by identifying data quality issues, troubleshooting inconsistencies, and implementing improvements to enhance dashboard accuracy and usability.
- Actively involved in delivering analytics solutions that enabled better decision-making and strengthened PETRONAS Upstream’s focus on data security, assurance, and control reporting.

Sigma Info Analytics Sdn Bhd, ***Data Analyst***

Jun 2023 - Oct 2024

- Built dynamic Power BI dashboards for PETRONAS Upstream, leveraging SQL Server integration to deliver cross-departmental reporting and actionable business insights.
- Managed end-to-end data integration and bridging from SQL Server and Oracle into Power BI, ensuring smooth data flow for accurate and reliable reporting.
- Performed data cleansing, transformation, and quality assurance to improve data consistency, integrity, and readiness for reporting and analytics.
- Provided technical support for Power BI services, troubleshooting report errors, resolving performance bottlenecks, and enhancing overall dashboard usability and user experience.
- Streamlined reporting operations through Power Automate, enabling automated data refreshes, notifications, and alerting processes to reduce manual intervention.
- Collaborated with business and technical stakeholders to maintain reporting efficiency and ensure dashboards aligned with operational and cross-functional requirements.
- Improved reporting reliability by identifying system and data issues early, supporting faster resolution and more stable dashboard performance.
- Contributed to a more efficient analytics environment by combining dashboard development, automation, and support, helping PETRONAS Upstream strengthen reporting delivery and decision-making.

Aventra Group, ***Data Engineer***

Jan 2023 - Nov 2023

- Managed Azure cloud infrastructure and Azure SQL Database environments, including performance monitoring, backup management, and high-availability configurations to ensure system reliability and operational continuity.
- Administered and optimized Azure SQL databases by performing query tuning, workload analysis, and database maintenance to enhance overall performance and stability.
- Implemented indexing strategies and SQL performance improvements that reduced query response times and increased efficiency for business-critical applications and reporting workloads.
- Strengthened platform resilience by supporting backup, recovery, and disaster recovery protocols for Azure SQL environments, ensuring data protection and business continuity readiness.
- Applied security controls and best practices across Azure environments, including access management, database security configurations, and compliance-focused measures to safeguard critical data assets.
- Collaborated with cross-functional teams to troubleshoot infrastructure and database-related issues, resolve incidents efficiently, and minimize service disruption.
- Supported deployment and continuous improvement initiatives by working with technical stakeholders to implement process enhancements, infrastructure updates, and operational best practices.
- Contributed to a more reliable and scalable cloud environment by combining database administration, security implementation, performance tuning, and infrastructure support.

Centre for Research in Data Science (CeRDaS), University Teknologi PETRONAS, ***Research Officer (Data Scientist)***

Jan 2022 - Dec 2022

- Built and deployed a Deep Learning model using Neural Networks to predict sand count production across PETRONAS wells, covering the full lifecycle from data preparation to production deployment.
- Designed and executed the end-to-end machine learning workflow, including business requirement gathering, problem formalization, data preprocessing, model development, testing, validation, and deployment.
- Prepared and transformed large datasets for model training by performing data cleaning, feature preparation, and quality checks to improve model reliability and predictive performance.
- Developed a web-based visualization dashboard hosted on Azure Web Services, enabling stakeholders to easily access predictive insights and monitor model outputs.
- Translated technical model results into clear and usable business insights, helping stakeholders make more informed decisions based on production forecasting.
- Collaborated with cross-functional teams to align machine learning solutions with operational needs, ensuring the model addressed real business challenges and delivered practical value.
- Supported deployment and accessibility of predictive analytics solutions by integrating machine learning outputs into a user-friendly dashboard environment.
- Contributed to innovation initiatives by combining data science, deep learning, cloud deployment, and visualization tools to deliver an end-to-end predictive solution for PETRONAS wells.

Downstream Business Excellence, PETRONAS, Kuala Lumpur, ***Data Analyst Intern***

Jun 2020 - Dec 2020

- Created interactive data visualizations and dashboards to support APC implementation decision-making, enabling clearer analysis of operational performance and business priorities.
- Applied AI-driven tools and analytical methods for budget forecasting and cost-saving analysis, helping identify opportunities to improve financial planning and operational efficiency.
- Synthesized and integrated data from multiple sources to produce comprehensive insights, supporting the development of strategic roadmaps and data-backed recommendations.
- Utilized business intelligence tools to transform complex datasets into clear, actionable visual reports for stakeholders and decision- makers.
- Analyzed trends, performance indicators, and cost drivers to support more informed planning, resource allocation, and implementation strategies.
- Collaborated with stakeholders to understand business needs and translate them into meaningful dashboards, reports, and analytical outputs.
- Improved visibility into project and budget performance by presenting data in a structured and accessible format for cross-functional teams.
- Contributed to strategic planning initiatives by combining data analysis, BI reporting, and forecasting techniques to support more effective decision-making.

EDUCATION

Universiti Teknologi PETRONAS

Bachelor of Technology (Hons.) • Information System

Major in Big Data Analytics. Minor in Corporate Management. CGPA: 3.53.

LICENSES & CERTIFICATIONS

Microsoft Certified: Data Analyst Associate
Issued Sep 2021

Analyzing Data with Microsoft Power BI; PETRONAS, KLCC
Issued Dec 2020

Certified IoT Security Professional
Universitas Brawijaya, Indonesia • Issued Dec 2019

SKILLS

- Data Engineering: Snowflake, Microsoft Fabric, Azure Data Factory, ETL/ELT, PySpark, Delta tables, Snowpipe, SFTP/API integration, external stages, stored procedures, incremental loading, deduplication
- Cloud & Infrastructure: Microsoft Azure, Terraform, Azure Blob Storage, Azure SQL, Azure DevOps, CI/CD
- Databases & SQL: Snowflake SQL, SQL Server, Oracle, MERGE/COPY INTO, CTEs, window functions
- Analytics & Visualisation: Power BI, DAX, Power Query, star schema and dimensional modelling
- CI/CD & Source Control: Azure DevOps, GitHub, Git, Azure Pipelines, pull requests, deployment support
- Programming & Other: SQL, Python, JavaScript, HTML, CSS, Power Automate, machine learning, data governance, Agile/Scrum